

# **Music & AI Hackathon**

11.09.2026 - 13.09.2026

## Challenge 1 - TOURISM

# Experience Sound & Nature

Austria is known by the world for music and culture. Being placed in the Alps some of the most scenic landscapes are enjoyed by 48.17 Mio people every year<sup>1</sup>. The National Parks like the Nationalpark Hohe Tauern (you are enjoying now) let visitors connect with nature in a meaningful way.

### Challenge:

How can Austria's natural soundscapes be turned into an interactive experience that helps people feel, understand and value natural landscapes?

### Challenge Questions:

1. How can sounds from nature be recorded, mixed or transformed into an experience people can interact with?
2. How can sound help tell stories about landscape, seasons, weather, tourism and sustainability?
3. How can the result encourage people to treat nature with more care?

**Available:** tourism context, natural sound inspiration, landscape and sustainability themes.

**Available tools:** Handheld recorders (5 sets) for your adventure

<sup>1</sup><https://www.statistik.at/en/statistics/tourism-and-transport/tourism/accommodation/arrivals-overnight-stays>

## Challenge 2 - Instruments & Sounds

# AI Instruments & Beyond Music

Creating music is no longer limited to traditional instruments. Small hardware, MIDI controllers, apps and trainable audio tools make it possible to build strange, playful and personal instruments. Roland Project LYDIA and Neutone Morpho plugin open the door to sounds that can react, change and behave in unexpected ways. But even as Music is a technology endeavor, live performances and people enjoying music together with the artists playing them has been more popular than ever.<sup>1</sup> And AI is certainly key to accelerating that endeavor.

### Challenge:

How can new instruments or sound objects be created by combining hardware, MIDI and experimental audio tools - usable in live settings? Or how can AI/Machine Learning effectively enhance artists' creativity?

### Challenge Questions:

1. How can a performer control sound in a way that feels physical and expressive?
2. How can everyday sounds, voices or objects become part of a new instrument?
3. Can the result be more than music, such as an installation, toy, educational tool, tools for well being, interface or performance object?

**Available tools:** AIRA Compact Series, MIDI instruments, Project LYDIA, Morpho plugin.

**Available Softwares:** PureData, or the programming environment you usually use.

<sup>1</sup> <https://news.livenationentertainment.com/news/live-nation-entertainment-reports-second-quarter-2026-results>

## Challenge 3 - Mobility

# Sound & Driving

Driving by car is the most popular form of mobility around the world<sup>1</sup>. A majority of time is spend in the car<sup>2</sup> and Music makes this time a very enjoyable moment for many people. A car already senses a lot: movement, pressure, position, humidity, current, nearby objects and the space around it. At the same time, driving depends heavily on attention, comfort and small unconscious actions. How can sound become a subtle way for the car to respond to the driver and the environment.

### Challenge:

How can a musical or auditory experience be created from a driver's movements, state and surroundings?

### Challenge Questions:

1. How can sensor data reveal what is happening inside or around the car?
2. How can sound support focus, calmness or awareness without distracting the driver?
3. How could the inside and outside of the vehicle become part of one sound experience?

**Available sensors and tools:** magnetic, pressure, humidity, current, electrostatic, position, millimeter-wave, Arduino

**Available Softwares:** Arduino IDE

1 <https://csh.ac.at/news/over-half-of-global-commutes-are-by-car-says-study/>

2 <https://www.volpe.dot.gov/news/how-much-time-do-americans-spend-behind-wheel>

## Challenge 4 - Game

# Game Sound & Spatial Interaction

Sound in Interaction with Space around the User as it's Avatar<sup>1</sup>. How would an experience look (or sound) like, that guides the User in the real world or a virtual game and provides an experience that causes reactions in it's host. We are looking for scenarios where Technology and sound pushes the possibilities of experiences that it's user might already have.

### Challenge:

Interactive audio experience in which music, spatial audio, or other dynamic sound behaviors respond to interaction, presence, or data? This audiokinetic challenge aims to get feedback on the software Wwise from it's teams.

### Challenge Questions:

1. How can sound change when the listener moves, enters a space, interacts with objects, or makes a choice? Consider the listener's relationship to the environment, such as object distance, direct interaction, or player-driven decisions.
2. How can spatial audio make an experience feel more alive or present?
3. How can the system remain musical, expressive, and coherent while reacting in real time?

**Available tools:** Wwise and Audiokinetic audio workflows. Participants may also explore game engine integrations, WAAPI, Wwise plug-ins, or Wwise Community Tools. Unity/Unreal Engine.

<sup>1</sup> [https://www.reddit.com/r/oculus/comments/auzzgo/in\\_1992\\_the\\_book\\_snow\\_crash\\_coined\\_the\\_term/](https://www.reddit.com/r/oculus/comments/auzzgo/in_1992_the_book_snow_crash_coined_the_term/)

## Challenge 5 - Academic Challenge

# Bodily Movement & Sounding Spaces

Your body becomes an instrument. Create a system where bodily movement becomes part of the music. Movement can control sound and graphics, but it can be more than that. This challenge invites you to explore that connection using AI and sensor technologies to turn movement into a rhythmic, melodic, or expressive layer of the music itself.

### Challenge:

How can a live performer's movement create sound and graphics in real time for a given topic?

### Challenge Questions:

1. How can "Movement" control sound? And can it be more than that?
2. Can you create an interactive environment where the performer's movements, captured by a standard webcam or a dedicated tracking camera, generate live sound and optionally also graphics in any synthesis or gaming environment?
3. How can you use phones, wearable sensors to capture gesture, rhythm, motion, or physical energy, and map this data to synthesis, VST instruments, samples, existing songs, or any audio recording?

**Available Tools:** webcams, projection, PA system and rehearsal space.

**Available Softwares:** SuperCollider, PureData, Godot Engine, Mediapipe or the programming environment you usually use.

## Challenge 6 - Accessibility

# Accessibility & Music

Music should be accessible to everyone. Inspired by tools such as the AlphaTheta Chordcat, this challenge explores how AI and new interfaces can lower barriers to music creation, performance, and learning for people with disabilities, beginners, or older adults.

### Challenge:

How can you create a more accessible and intuitive way for anyone to engage with music?

### Challenge Questions:

1. How can AI help people with visual, hearing, motor, or other disabilities participate in music creation and performance in new ways?
2. How can interfaces, sensors, voice, gesture, or other forms of interaction make musical tools easier and more intuitive to use for people of different ages and abilities?
3. How can AI reduce the learning curve for beginners and older adults, helping people with little or no musical or technical experience start creating music quickly and enjoyably?

**Available Tools:** AlphaTheta ChordCat (5 sets)

**Available Softwares:** The programming environment you usually use.